# Developer Report — Engineering Analysis Suite

## 1. Interesting engineering insight

The pipe-flow module shows how strongly pressure loss depends on flow rate. Because Darcy-Weisbach pressure loss contains the velocity-squared term, increasing flow rate can produce a much larger pressure drop even when pipe geometry stays unchanged. The pressure-drop-versus-flow-rate plot therefore helps explain hydraulic operating limits.

## 2. Technical challenge overcome

The main challenge was combining engineering calculations with a robust interactive web interface. I separated reusable calculations into an engineering.py module and used input validation so invalid values do not crash the application. I checked the pipe-flow, Fourier conduction, and Newton-cooling results against hand/analytical solutions before treating the interface as complete.

## 3. What I would add with more time

I would add minor-loss calculations for valves and fittings, more friction-factor correlations such as Colebrook-White, transient conduction for cases where the lumped-capacitance assumption is not valid, and richer rock/fluid data quality checks. I would also add automated unit conversion and downloadable reports.

## Verification highlights

| Module | Verification case | Expected result |
|---|---|---|
| Pipe flow | D=0.10 m, L=100 m, Q=0.005 $m^3/s$, water | v≈0.637 m/s; Re≈63,408; ΔP≈4.29 kPa |
| Conduction | k=0.60, A=10, ΔT=75 K, L=0.10 m | Q■=4500 W |
| Cooling | T0=100°C, $T_\infty$=25°C, target=40°C | t≈6.44 min |

**Submission placeholders:** GitHub repository URL: ____________________ Live Streamlit URL: ____________________